Communication Complexity

$f: X \times Y \longrightarrow \{0,1\}$

X, Y finite. Can encode f using a $|X| \times |Y|$ matrix $M(f)$, 0-1 matrix

(Informal) Goal: Alice has $x \in X$,
Bob has $y \in Y$

We want to evaluate $f(x,y)$ with minimum communication.

Many variations. 1-way, 2-way, etc.
used elsewhere, e.g. sketching lower bounds (Clarkson & Woodruff and others)

Non-Deterministic Comm Comp

1. Alice & Bob see $f : X \times Y \to \{0,1\}$ and hence $M(f)$. They can collaborate/comm/comp.

2. They separate. They receive x, y resp.

3. "Prover" sees x & y and sends to both the same "hint" or "proof" : $i \in \{1, \dots, t\}$

4. Bob and Alice accept or reject the proof.

<u>Succeeds</u> if $\exists$ mutually acceptable "proof" whenever $f(x,y) = 1$

and $\exists$ a non mutually accept. proof when $f(x,y) = 0$

Ex

$M(f) =$